Name_____ Date_____ Period_____

LESSON 5-6 · HOMEWORK PACKET · SUPPORTED · SESSION 1 · PAGES 1–2

Drawing Nets of Solids

Maryland Standard 6.G.A.4 · Learning target – I can draw the net of a solid by unfolding its faces onto a flat surface.

REMEMBER

- A **net** is a flat pattern that folds into a solid. Each face appears **once**, with no overlaps.
- **Rectangular prism**: 6 rectangles. **Square pyramid**: 1 square + 4 triangles. **Triangular prism**: 2 triangles + 3 rectangles.

WORKED EXAMPLE

Plan a net for a box 5 cm × 3 cm × 2 cm.

Step 1 Faces come in pairs: 5 × 3, 5 × 2, 3 × 2

Step 2 Draw the 5 × 3 base. Attach a 5 × 2 face to each long edge.

Step 3 Attach a 3 × 2 face to each short edge. Add the 5 × 3 top to one 5 × 2 face.

Answer **6 rectangles, two of each size**

WORD BANK

net – a flat pattern that folds into a solid

face – a flat side of a solid

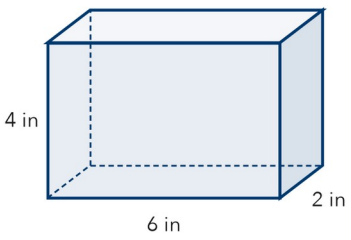
rectangular prism – a box shape

square pyramid – a square base + 4 triangles

PRACTICE

show your work · label units

1. Draw a net for this box. Label the length and width of every face.



Step 1 Draw the bottom face first.

Step 2 Attach a face to each edge of the bottom.

Step 3 Add the top. Label every face.

Draw the net here.

Face sizes_____

2. How many of each shape are in the net?

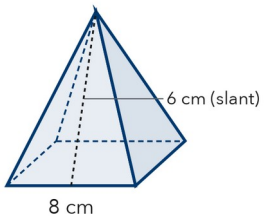
a. cube: squares ___

b. square pyramid: triangles ___

c. triangular prism: rectangles ___

d. pentagonal pyramid: triangles ___

3. The net of this square pyramid has a square and 4 triangles.



Step 1 Square: side × side.

Step 2 Triangle: ½ × base × height.

a. Area of the square _____ cm²

b. Area of one triangle _____ cm²

All 4 triangles_____ cm²

4. Cereal Box. A cereal box is 8 in long, 3 in wide, and 12 in tall. Its net has **3 pairs** of matching faces. List the size of each pair.

Face pairs_____

5. Will It Fold? Jamal draws 6 squares in one long row and says it is a net for a cube. Is he right? Explain.

Frame *Jamal is ___ because a row of 6 squares ___.*

_____

_____

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Open an empty cereal box along its edges. Show someone its net.

Name_____ Date_____ Period_____

LESSON 5-6 · HOMEWORK PACKET · SUPPORTED · SESSION 2 · PAGES 3–4

Reading Nets and Solids

Maryland Standard 6.G.A.4 · Learning target – I can name the solid a net makes by reading the shapes in the net.

REMEMBER

- **Prism**: 2 matching bases + rectangles. **Pyramid**: 1 base + triangles that meet at a point.
- The **base shape** names the solid (triangular, square, pentagonal …).

WORKED EXAMPLE

A net has 1 hexagon and 6 triangles.

Step 1 One base with triangles around it means a pyramid.

Step 2 The base is a hexagon.

Answer **hexagonal pyramid**

WORD BANK

prism – 2 matching bases + rectangles

pyramid – 1 base + triangles that meet at a point

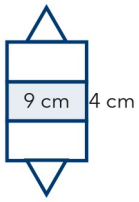
base shape – the shape that names the solid

edge – where two faces meet

PRACTICE

show your work · label units

1. Look at the net.



Step 1 Count the triangles and the rectangles.

Step 2 Two matching triangles + rectangles make a prism.

a. Solid: _____

b. Number of faces: ___

2. Name the solid each net makes.

Step 1 Two matching shapes: prism. One base + triangles: pyramid.

Step 2 Use the base shape in the name.

a. 1 rectangle + 4 triangles: _____

b. 2 hexagons + 6 rectangles: _____

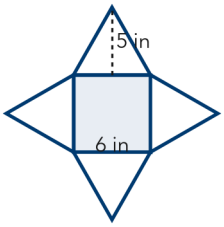
c. 2 triangles + 3 rectangles: _____

d. 1 pentagon + 5 triangles: _____

e. 6 squares: _____

f. 2 pentagons + 5 rectangles: _____

3. This net folds into a square pyramid.



Step 1 Base: 6 × 6.

Step 2 One triangle: ½ × 6 × 5. Then multiply by 4.

a. Area of the base: _____ in²

b. Area of all 4 triangles: _____ in²

4. **Mystery Box.** A net has 2 **squares** (4 cm on each side) and 4 **rectangles** (4 cm × 9 cm). Name the solid. Give its length, width, and height.

Answer_____

TEST PRACTICE

choose the best answer

5. A triangular prism has 5 faces. How many edges does it have?

Step 1 Count the edges on each triangle end.

Step 2 Add the edges that join the two ends.

A 5 **B** 6 **C** 9 **D** 12

6. How many faces does a hexagonal pyramid have?

A 6 **B** 7 **C** 8 **D** 12

CHECK YOURSELF	□ I can do this on my own	□ I need a little help	□ I have a question about #___

AT HOME Find a box at home. Tell someone what shapes its net would have and how many of each.